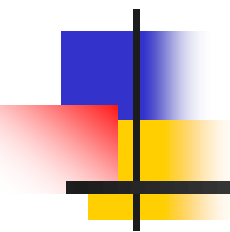




Indian Institute of Information Technology, Surat

भारतीय सूचना प्रौद्योगिकी संस्थान, सूरत
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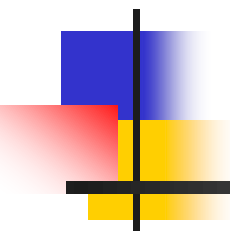
EC 503: Image Processing and Computer Vision

By:

Dr. Hemant S. Goklani

ECE Department

IIIT, Surat



UNIT 4

Image Segmentation and Image Representation and Description

UNIT4_1: Image Segmentation

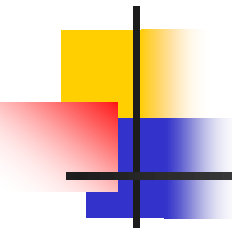
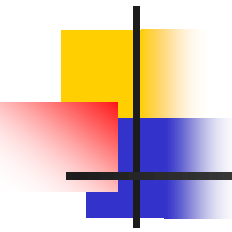


Image Segmentation

- Image segmentation refers to the process of partitioning an image into groups of pixels which are homogeneous with respect to some criterion
- Different groups must not intersect with each other, and adjacent groups must be heterogeneous
- Segmentation algorithms are area oriented instead of pixel-oriented
- The result of segmentation is the splitting up of the image into connected areas
- Thus segmentation is concerned with dividing an image into meaningful regions



Classification of Image Segmentation Techniques

1. **Local segmentation:** Local segmentation deals with segmenting sub-images which are small windows on a whole image
2. **Global segmentation:** Global segmentation is concerned with segmenting a whole image

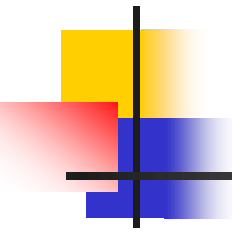
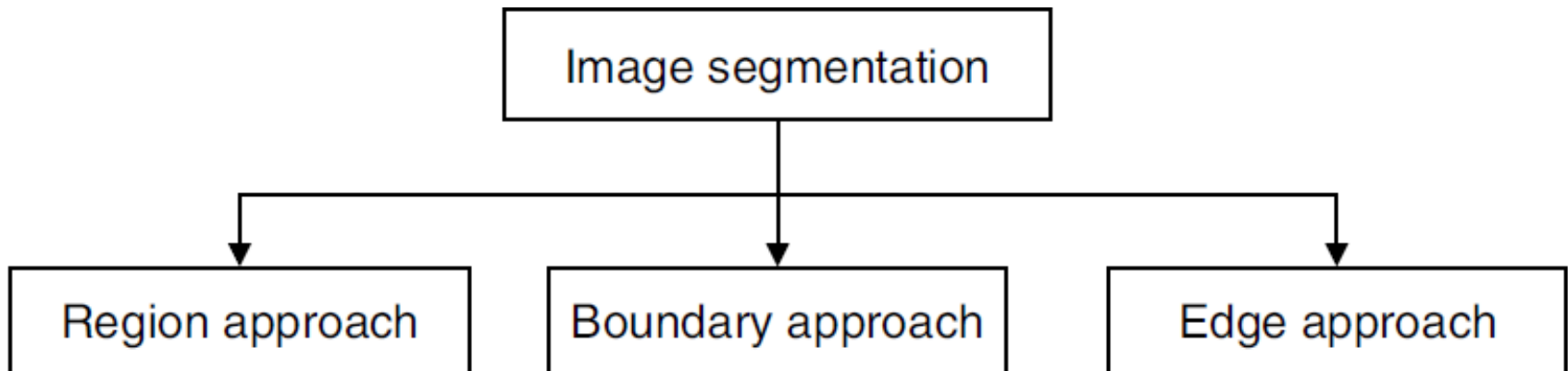
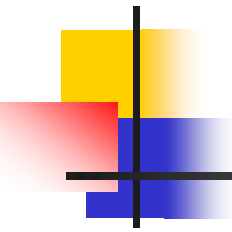


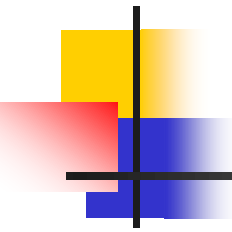
Image Segmentation





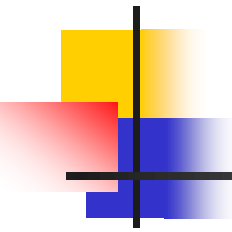
Region approach to image segmentation

- Regions in an image are a group of connected pixels with similar properties
- In this approach, each pixel is assigned to a particular object or region



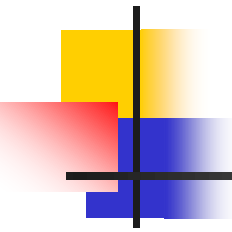
Region growing

- Region growing is an approach to image segmentation in which neighbouring pixels are examined and added to a region class if no edges are detected
- This process is iterated for each boundary pixel in the region
- If adjacent regions are found, a region-merging algorithm is used in which weak edges are dissolved and strong edges are left intact
- Region growing requires a seed to begin with. Ideally, the seed would be a region, but it could be a single pixel
- A new segment is grown from the seed by assimilating as many neighbouring pixels as possible that meet the homogeneity criterion



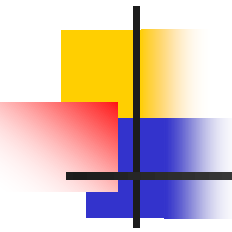
Region growing

- The resultant segment is then removed from the process
- A new seed is chosen from the remaining pixels. This continues until all pixels have been allocated to a segment
- As pixels are aggregated, the parameters for each segment have to be updated
- The resulting segmentation could depend heavily on the initial seed chosen and the order in which neighbouring pixels are examined
- The selection of homogeneity criteria in image growing depends not only on the problem under consideration but also on the type of image to be segmented



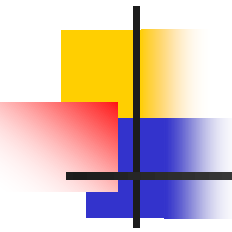
Region growing

- Region-growing algorithms vary depending on the criteria used to decide whether a pixel should be included in the region or not, the connectivity type used to determine neighbours, and the strategy used to visit neighbouring pixels
- Region growing offers several advantages over conventional segmentation techniques
- The borders of regions found by region growing are perfectly thin and connected
- The algorithm is also very stable with respect to noise



Region Splitting

- Region splitting is a top-down approach
- It begins with a whole image and divides it up such that the segregated parts are more homogenous than the whole
- Splitting alone is insufficient for reasonable segmentation, as it severely limits the shapes of segments
- Hence, a merging phase after the splitting phase is always desirable, which is termed as the split-and-merge algorithm



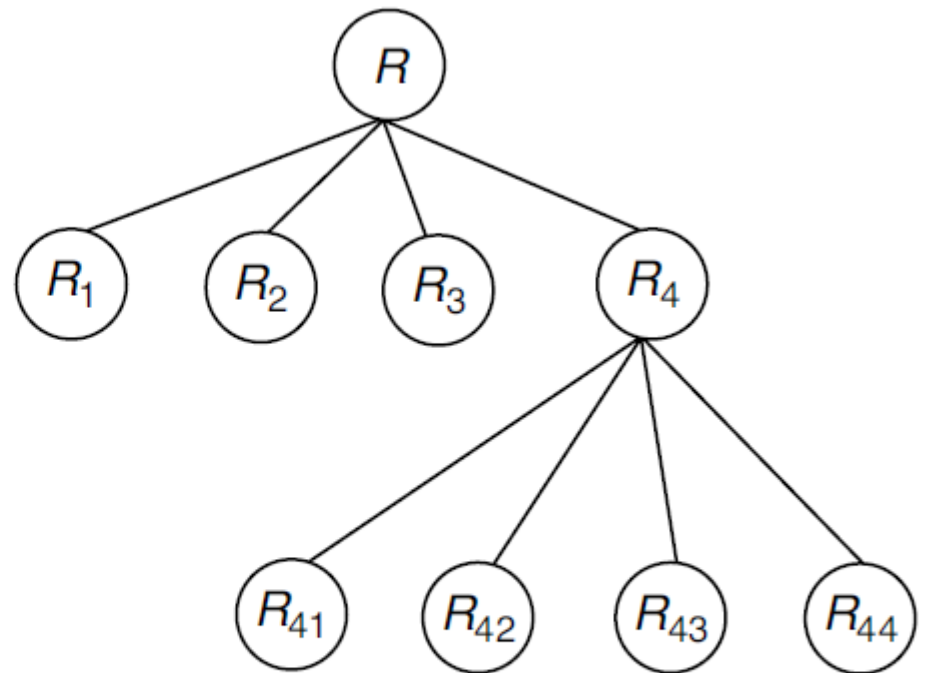
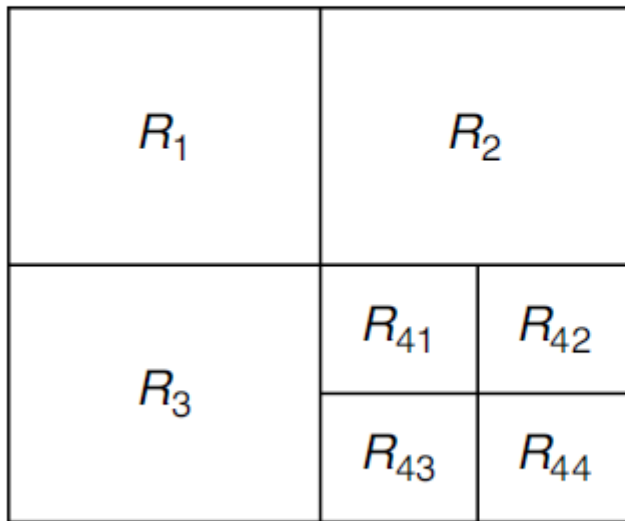
Region Splitting and Merging

- Region splitting and merging is an image-segmentation technique that takes spatial information into consideration
- The region-splitting-and-merging method is as follows:

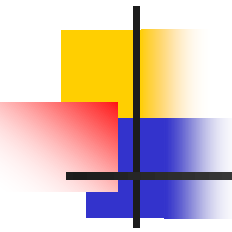
Splitting

- Let R represent the entire image
 1. Select a predicate P
 2. Split or subdivide the image successively into smaller and smaller quadrant regions
- The splitting technique has a convenient representation in the form of a structure called a quadtree
- In a quadtree, the root of the tree corresponds to the entire image and each node corresponds to a subdivision

Region Splitting and Merging



(a) Splitting of an image (b) Representation by a quadtree



Region Splitting and Merging

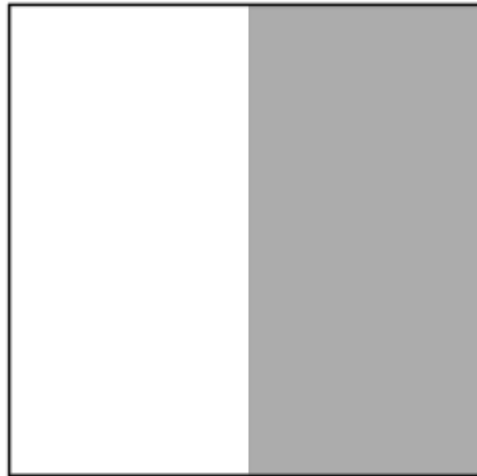
- The final partition is likely to contain adjacent regions with identical properties
- This drawback may be fixed by applying merging, and merging only adjacent regions whose combined pixels satisfy the predicate P

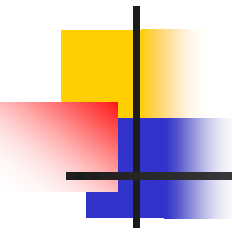
Merging

- Merge any adjacent regions that are similar enough
- The procedure for split and merge algorithm is given below:
 1. Start with the whole image
 2. If the variance is too large, break it into quadrants
 3. Merge any adjacent regions that are similar enough
 4. Repeat steps (2) and (3) iteratively until no more splitting or merging occurs

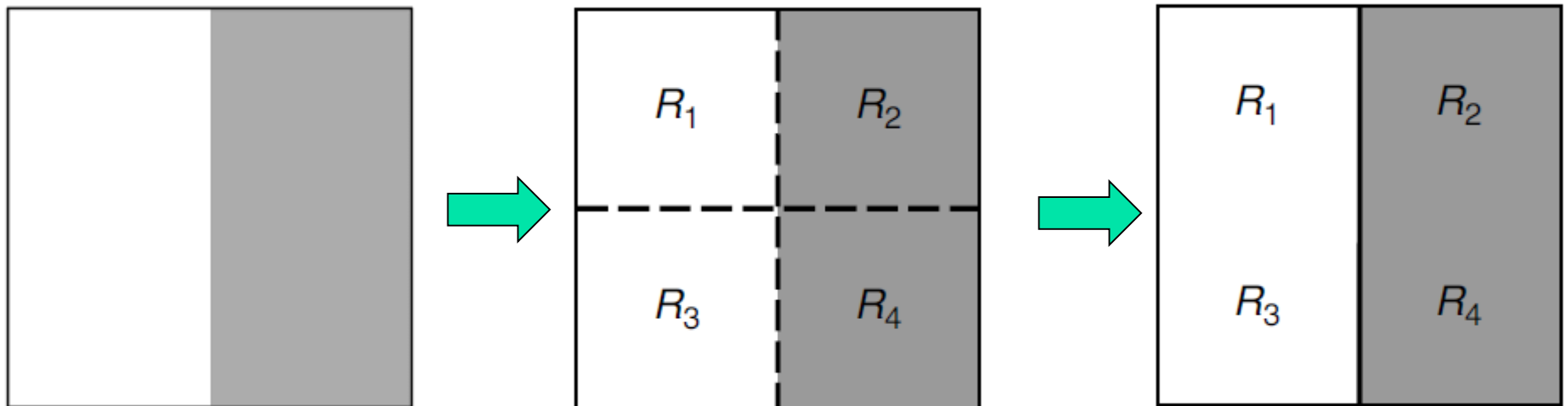
Example 1

- Apply the split-and-merge technique to segment the following image



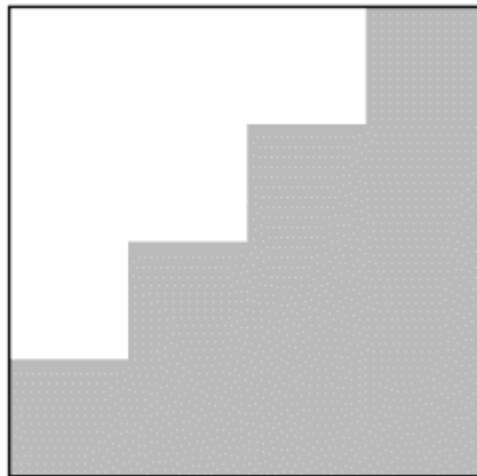


Solution

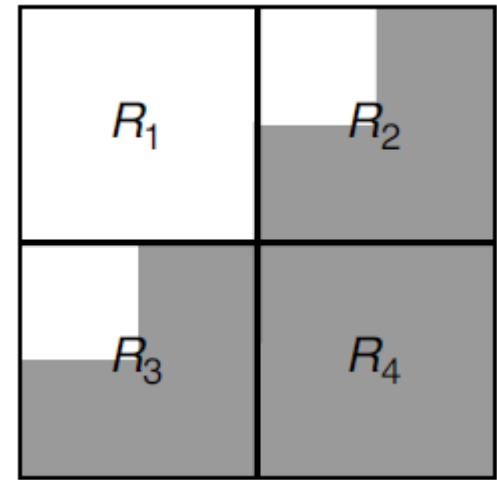
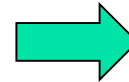
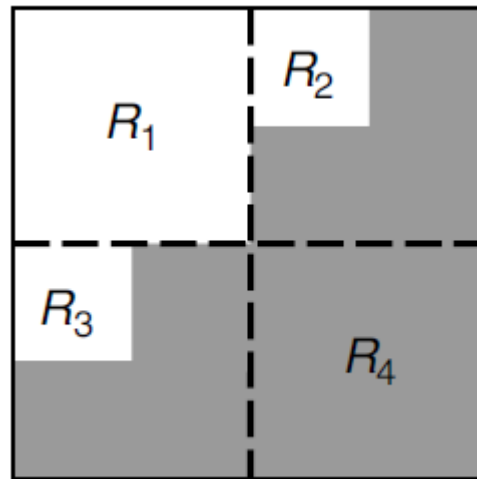
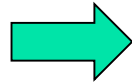
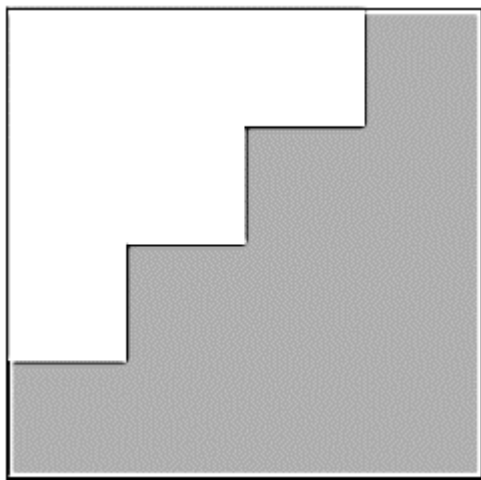


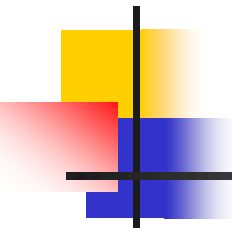
Example 2

- Apply the split-and-merge technique to segment the following image

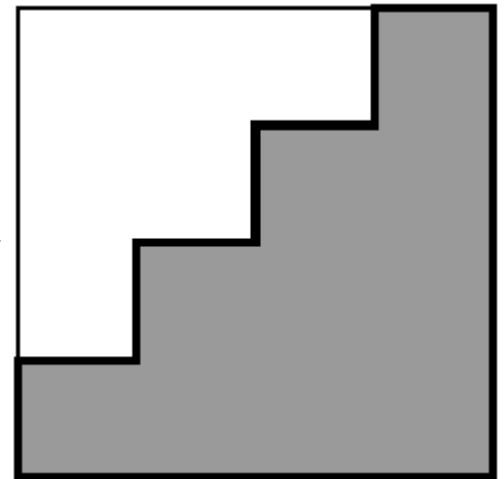
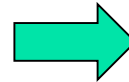
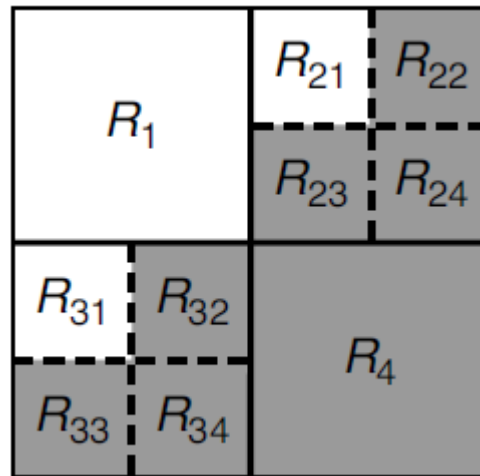
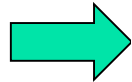
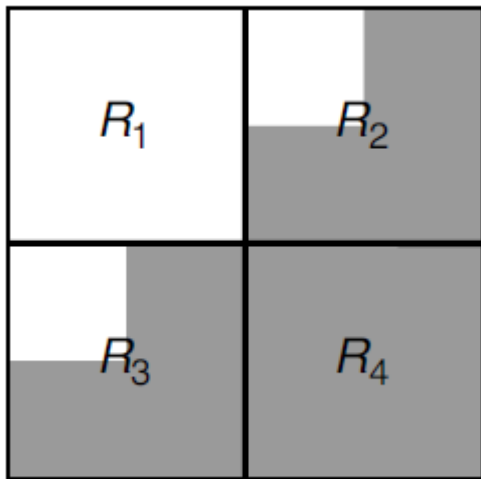


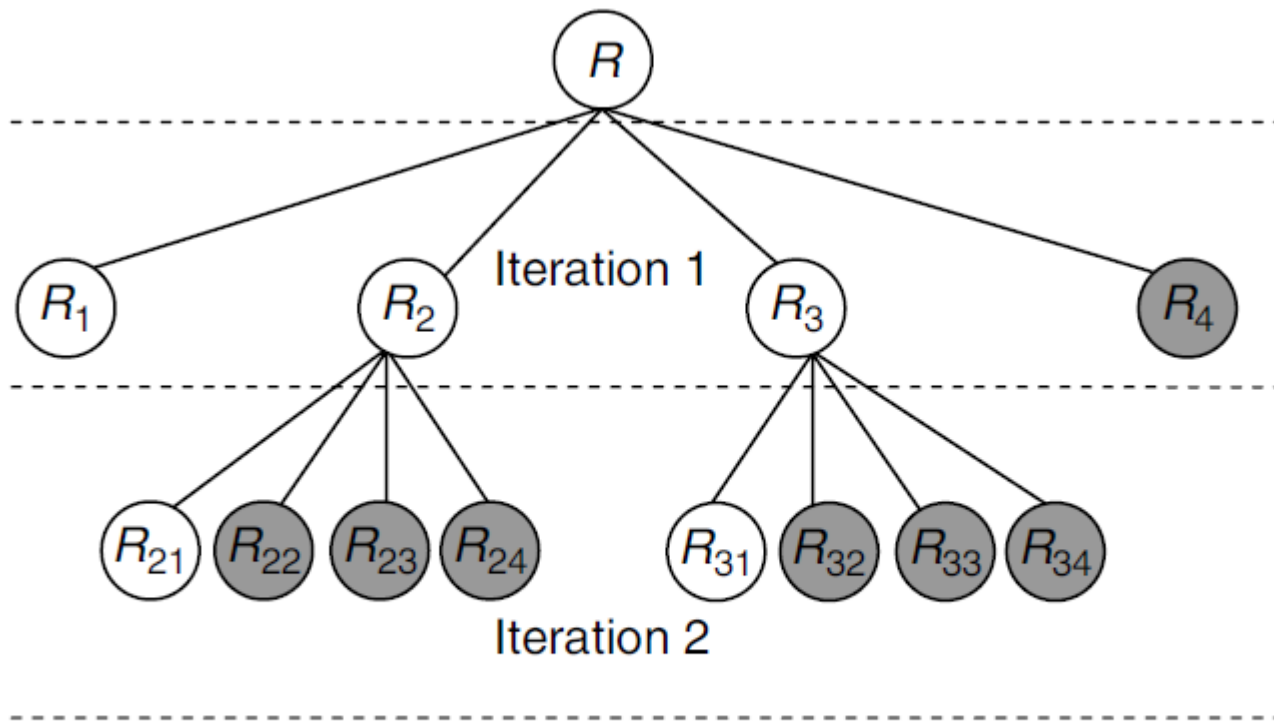
Solution



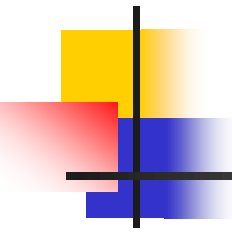


Solution



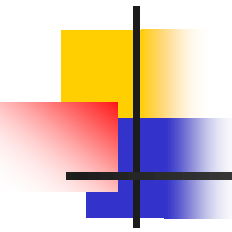


Quadtree representation



Clustering Techniques

- Clustering technique attempts to access the relationships among patterns of the data set by organizing the patterns into groups or clusters such that patterns within a cluster are more similar to each other than patterns belonging to different clusters
- Clustering refers to the classification of objects into groups according to certain properties of these objects
- An attempt is made to extract a feature vector from local areas in the image
- A standard procedure for clustering is to assign each pixel to the class of the nearest cluster mean
- Clustering methods can be divided into two categories
 1. Hierarchical and
 2. Partitional

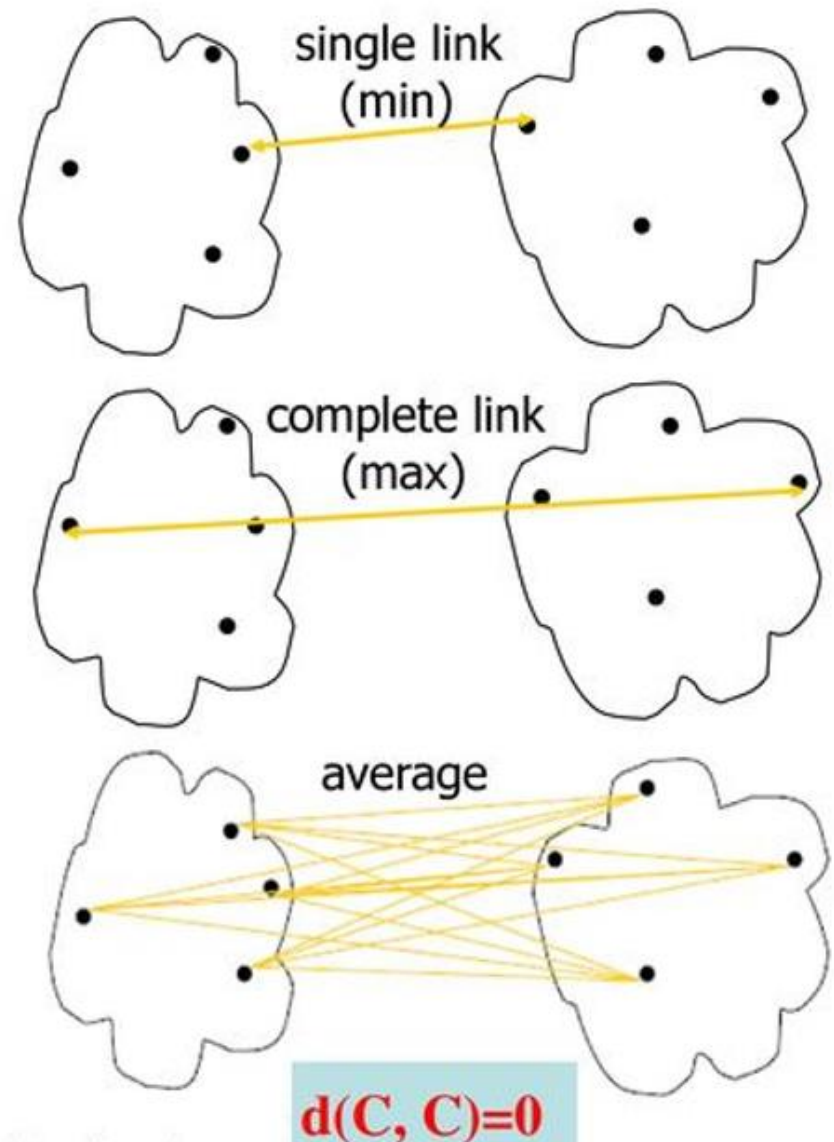


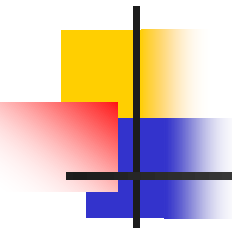
Hierarchical Clustering

- Hierarchical clustering techniques are based on the use of a proximity matrix indicating **the similarity between every pair of data points to be clustered**
- The end result is a tree of clusters representing the nested group of patterns and similarity levels at which groupings change
- The resulting clusters are always produced as the internal nodes of the tree, while the root node is reserved for the entire dataset and leaf nodes are for individual data samples
- The clustering methods differ in regard to the rules by which two small clusters are merged or a large cluster is split
- The two main categories are:
 1. Agglomerative and
 2. Divisive

Cluster Distance Measures

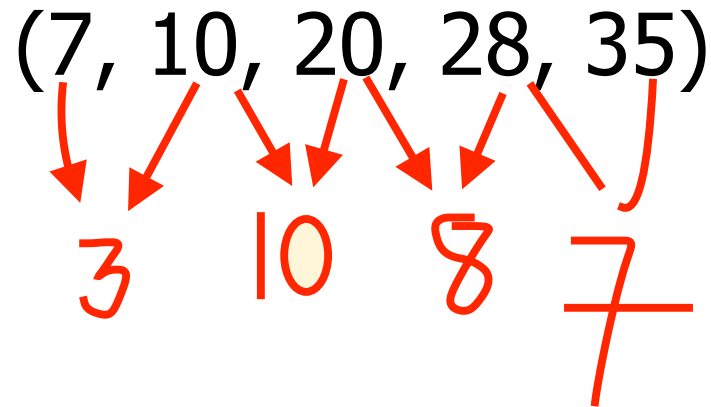
- **Single link:** smallest distance between an element in one cluster and an element in the other, i.e.,
$$d(C_i, C_j) = \min\{d(x_{ip}, x_{jq})\}$$
- **Complete link:** largest distance between an element in one cluster and an element in the other, i.e.,
$$d(C_i, C_j) = \max\{d(x_{ip}, x_{jq})\}$$
- **Average:** avg distance between elements in one cluster and elements in the other, i.e.,
$$d(C_i, C_j) = \text{avg}\{d(x_{ip}, x_{jq})\}$$





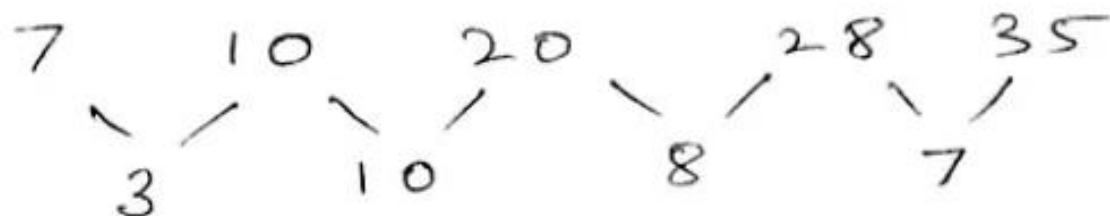
Example

- Do the clustering for the following using single link method

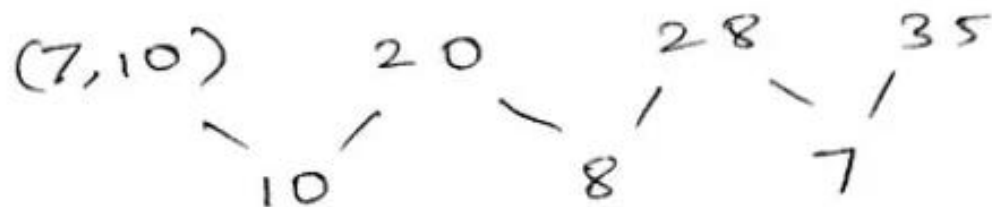


Single Linkage

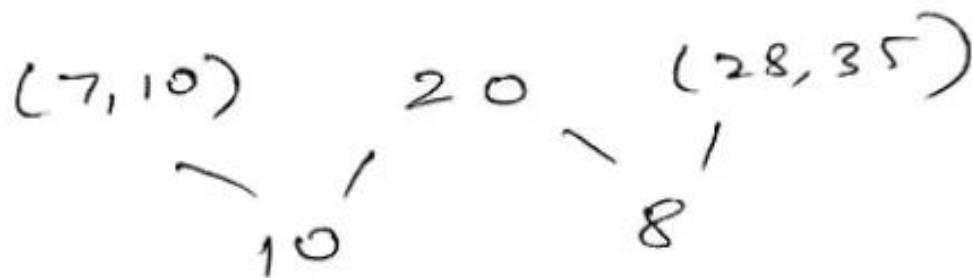
①



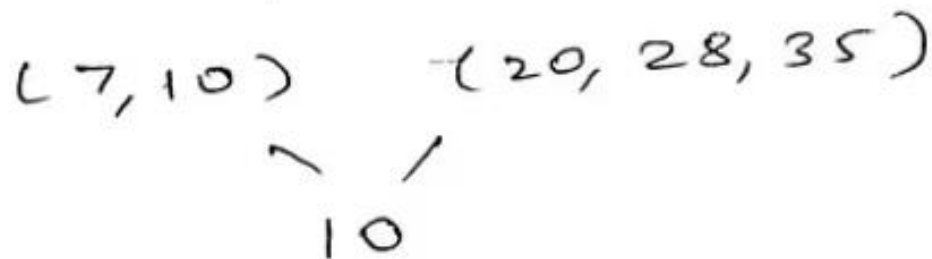
②

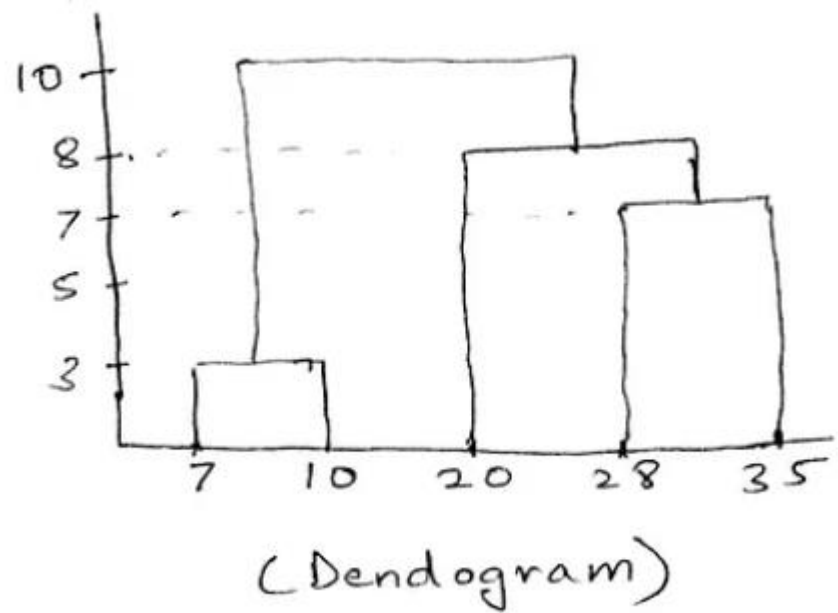


③



④





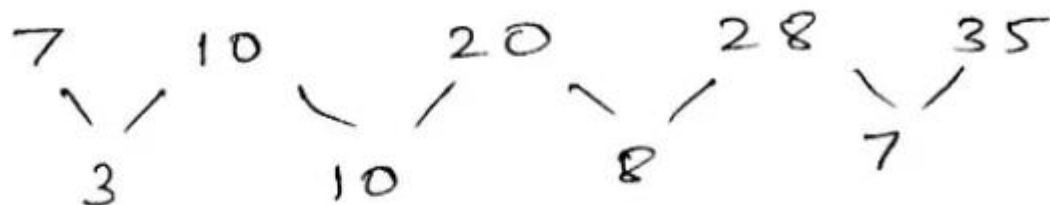
Using single linkage two clusters are formed

Cluster 1 : (7,10)

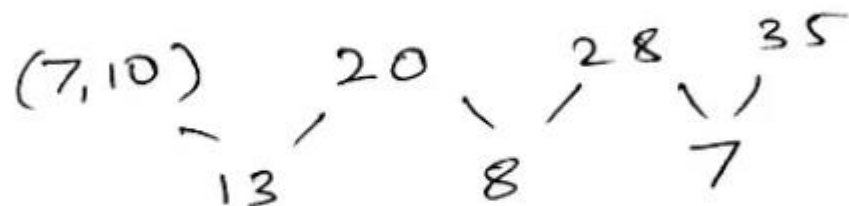
Cluster 2 : (20,28,35)

Complete Linkage

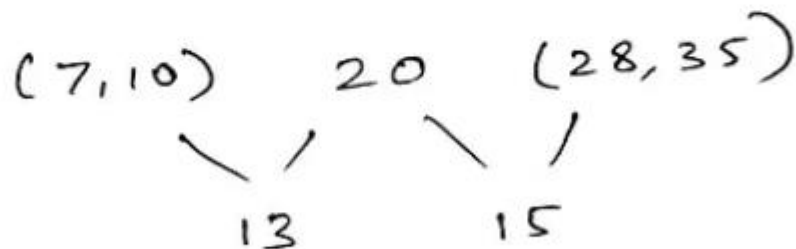
①



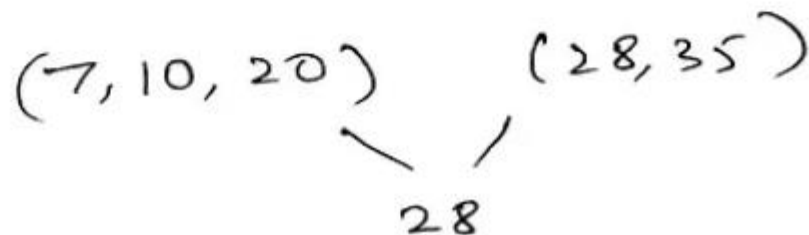
②

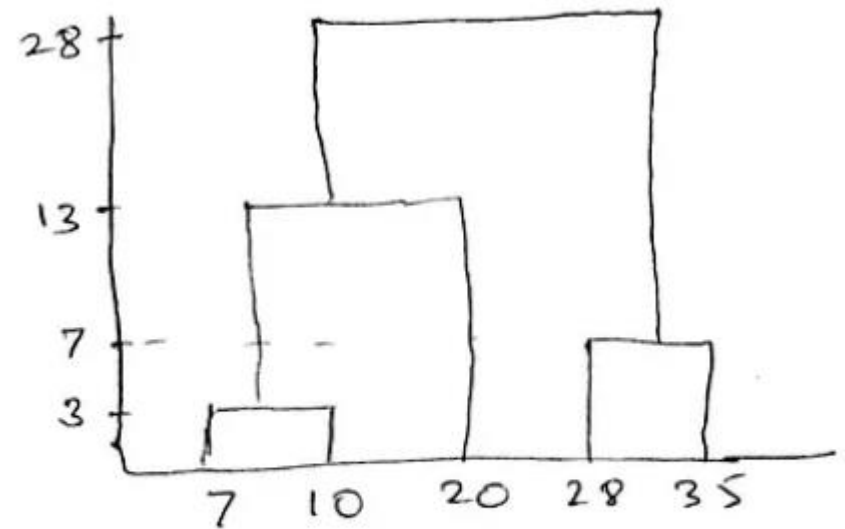


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④

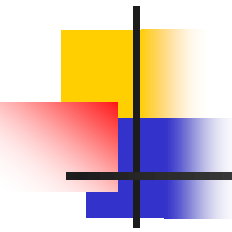




Using complete linkage two clusters are formed : (Dendrogram)

Cluster 1 : (7,10,20)

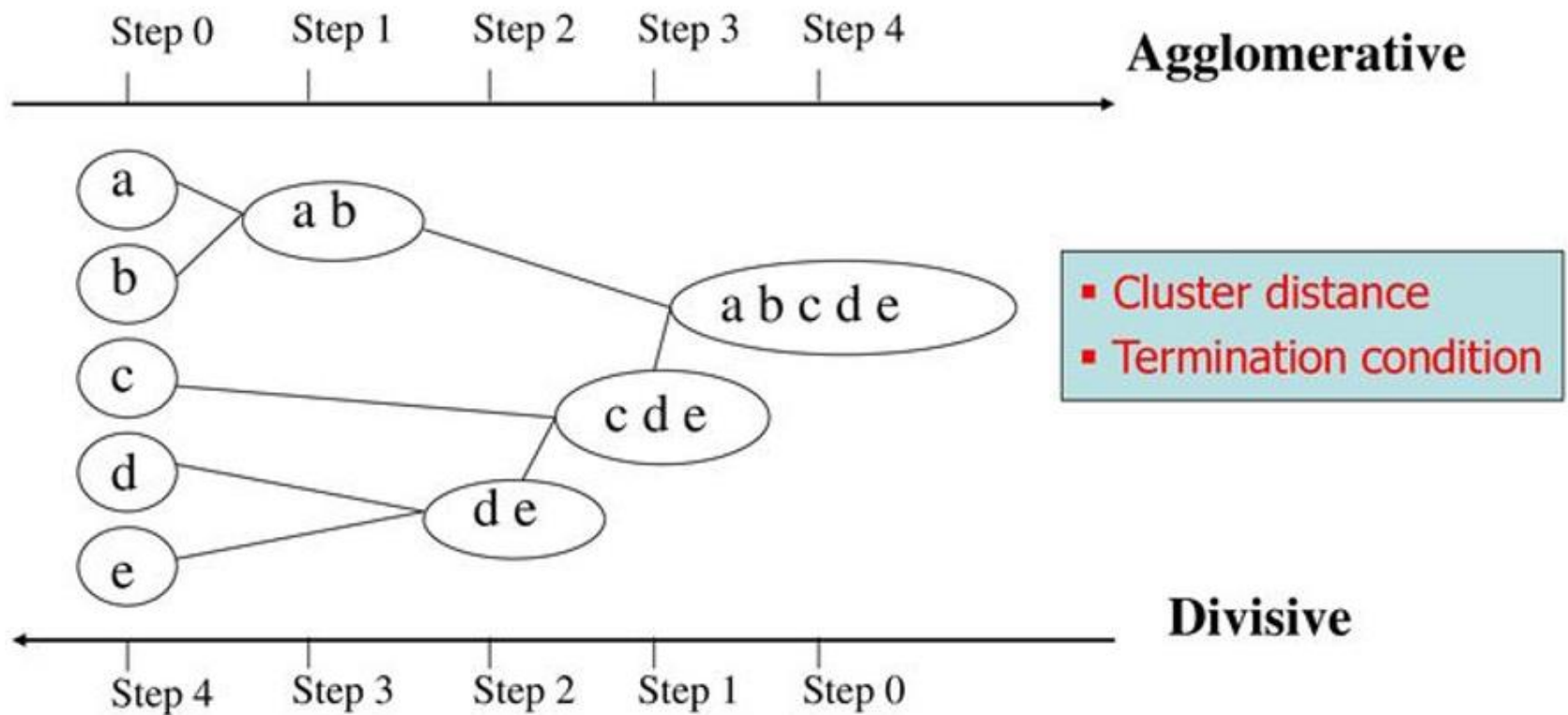
Cluster 2 : (28,35)

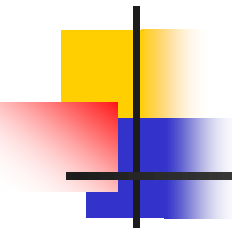


Divisive Algorithm

- Divisive clustering begins with the entire dataset in the same cluster, followed by iterative splitting of the dataset until the single-point clusters are attained on leaf nodes
- It follows a reverse clustering strategy against agglomerative clustering. On each node, the divisive algorithm conducts a full search for all possible pairs of clusters for data samples on the node
- Some of the hierarchical algorithms include COBWEB, CURE and CHAMELEON

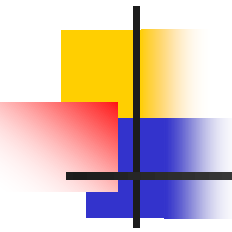
- Illustrative Example: Agglomerative vs. Divisive
Agglomerative and divisive clustering on the data set $\{a, b, c, d, e\}$





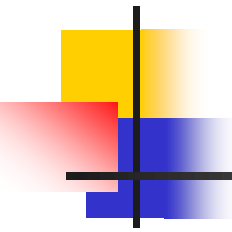
Partitional Clustering

- Partition-based clustering uses an iterative optimisation procedure that aims at minimising an objective function f , which measures the goodness of clustering
- Partition-based clusterings are composed of two learning steps—the partitioning of each pattern to its closest cluster and the computation of the cluster centroids
- A common feature of partition-based clusterings is that the clustering procedure starts from an initial solution with a known number of clusters
- The cluster centroids are usually computed based on the optimality criterion such that the objective function is minimized



Partitional Clustering

- Partitional algorithms are categorised into partitioning relocation algorithms and density-based partitioning
- Algorithms of the first type are further categorised into probabilistic clustering, K-medoids, and K-means
- The second types of partitional algorithms, called density-based partitioning, include algorithms such as DBSCAN, OPTICS DBCLAS, DENCLUE



K-means Clustering

- The K-means method is the simplest method in unsupervised classification
- The clustering algorithms do not require training data
- K-means clustering is an iterative procedure
- The K-means clustering algorithm clusters data by iteratively computing a mean intensity for each class and segmenting the image by classifying each pixel in the class with the closest mean
- **Limitations:**
- The drawback the K-means algorithm is that the number of clusters is fixed, once K is chosen and it always returns K cluster centres

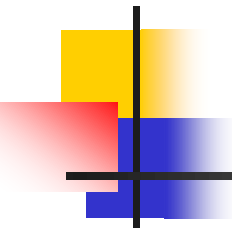
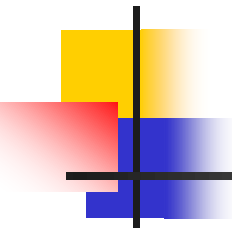


Image Segmentation based on Thresholding

- Thresholding techniques produce segments having pixels with similar intensities
- Thresholding is a useful technique for establishing boundaries in images that contain solid objects resting on a contrasting background
- The thresholding technique requires that an object has homogenous intensity and a background with a different intensity level
- Such an image can be segmented into two regions by simple thresholding

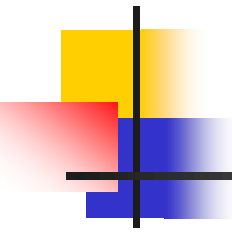


Global Thresholding

- In global thresholding, a threshold value of θ is chosen and the following condition is imposed

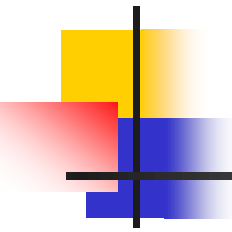
$$f(m, n) = \begin{cases} 1 & \text{if } f(m, n) \geq \theta \\ 0 & \text{else} \end{cases}$$

- Binarisation algorithm
- The value of θ has to be selected in an optimal manner
- Global thresholding will suffer when pixels from different segments overlap in their use of intensities



Adaptive Thresholding

- Global thresholding, or fixed thresholding, works well if the objects of interest have a reasonably uniform interior gray level and rest on a background of unequal but a relatively uniform gray level
- In many cases, the background gray level is not constant, and object contrast varies within an image
- In such cases, a threshold that works well in one area might not work well in other areas of the image
- In these cases, it is convenient to use a threshold gray level that is a slowly varying function of position in the image

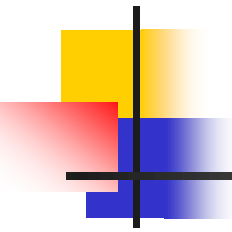


Histogram-Based Threshold Selection

- An image containing an object on a contrasting background has a bimodal gray-level histogram
- The two peaks correspond to the relatively large number of points inside and outside the object
- The dip between the peaks corresponds to the relatively few points around the edge of the object
- This dip is commonly used to establish the threshold gray level
- The histogram is the derivative of the area function for an object whose boundary is defined by thresholding:

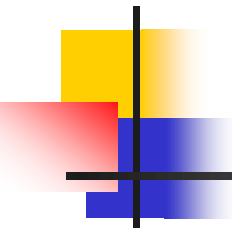
$$H(D) = -\frac{d}{dD} A(D)$$

- where D represents the gray level, $A(D)$ represents the area of the object obtained by thresholding at gray level D , and $H(D)$ represents the histogram



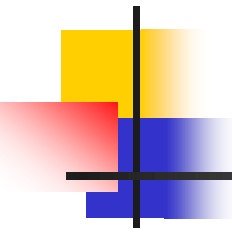
Histogram-Based Threshold Selection

- Increasing the threshold from D to $D + \Delta D$ cause only a slight decrease in area if D is near the dip of the histogram
- Therefore, placing the threshold at the dip in the histogram minimises the sensitivity of the area measurement to small errors in threshold selection
- If the image containing the object is noisy and not large, the histogram itself will be noisy
- Unless the dip is uncommonly sharp, the noise can make its location obscure, or at least unreliable from one image to the next
- This effect can be overcome to some extent by smoothing the histogram using either a convolution filter or the curve-fitting procedure



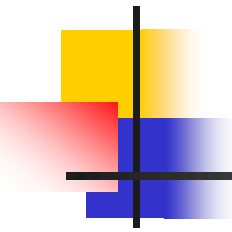
Limitation of Thresholding Technique

- The main limitation of thresholding techniques is that in its simplest form, **only two classes are generated and it cannot be applied to multi-channel images**
- A thresholding technique does not take into account the spatial characteristics of an image
- This causes it to be sensitive to noise and intensity inhomogeneities



EDGE-BASED SEGMENTATION

- Edge-based segmentation exploits spatial information by detecting the edges in an image
- Edges correspond to discontinuities in the homogeneity criterion for segments
- Edge detection is usually done with local linear gradient operators such as Prewitt, Sobel and Laplacian filters
- These operators work well for images with sharp edges and low amounts of noise
- The detected boundaries using these operators may not necessarily form a set of closed connected curves, so some edge linking may be required



CLASSIFICATION OF EDGES

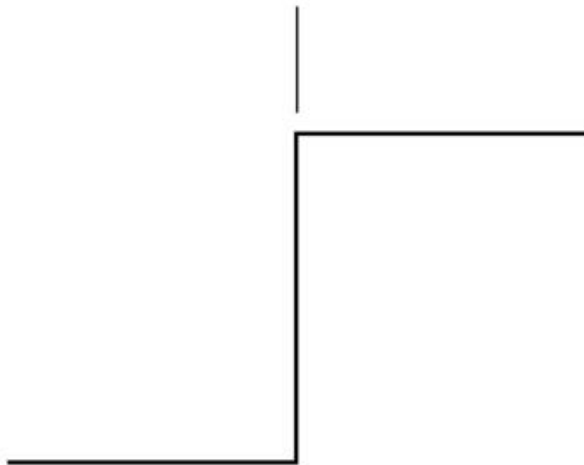
- Points in an image where brightness changes abruptly are called edges or edge points
- Edges are significant local changes of intensity in an image
- Edges are the boundaries between segments
- Edges are a very important portion of the perceptual information content in an image
- Edges can be broadly classified into
 - i. Step edge
 - ii. Line edge
 - iii. Ramp edge and
 - iv. Roof edge

CLASSIFICATION OF EDGES

Model of an ideal digital edge



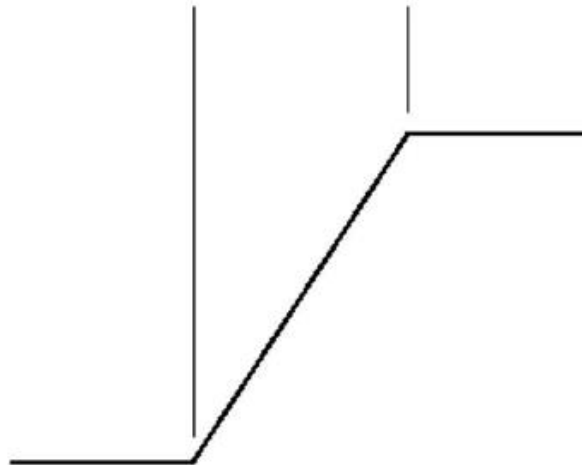
Gray-level profile
of a horizontal line
through the image



Model of a ramp digital edge



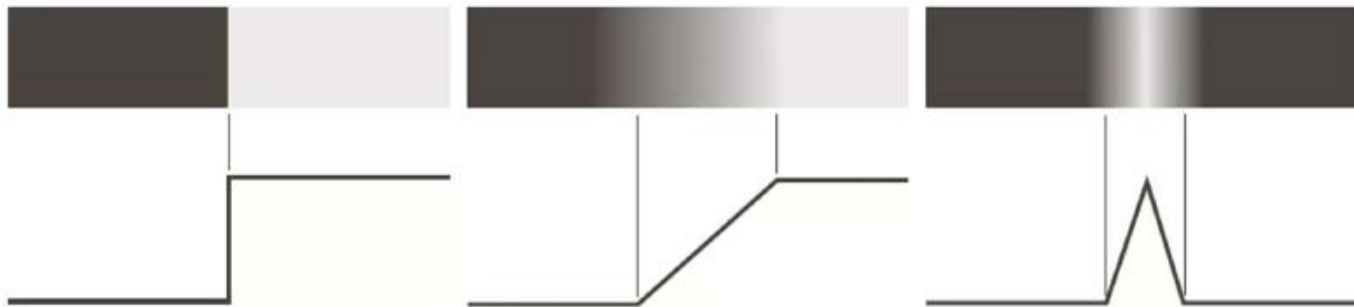
Gray-level profile
of a horizontal line
through the image



a b

FIGURE 10.5
(a) Model of an ideal digital edge.
(b) Model of a ramp edge. The slope of the ramp is proportional to the degree of blurring in the edge.

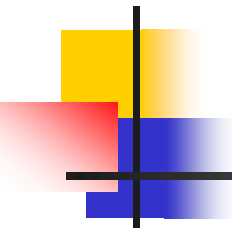
CLASSIFICATION OF EDGES



a b c

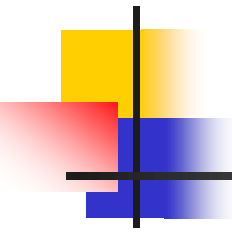
FIGURE 10.8

From left to right, models (ideal representations) of a step, a ramp, and a roof edge, and their corresponding intensity profiles.



EDGE DETECTION

- Edge detection is the process of finding meaningful transitions in an image
- The points where sharp changes in the brightness occur typically form the border between different objects
- These points can be detected by computing intensity differences in local image regions
- The edge-detection algorithm should look for a neighbourhood with strong signs of change
- Most of the edge detectors work on measuring the intensity gradient at a point in the image
- The purpose of edge detection is to identify areas of an image where a large change in intensity occurs, which are often associated with some physical boundary in the scene from which the image is derived



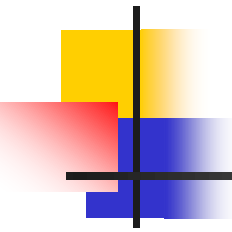
Gradient Operator

- A gradient is a two-dimensional vector that points to the direction in which the image intensity grows fastest
- The gradient operator ∇ is given by

$$\nabla = \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{bmatrix}$$

- If the operator ∇ is applied to the function f then

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$



Gradient Operator

- The two functions that can be expressed in terms of the directional derivatives are the gradient magnitude and the gradient orientation
- It is possible to compute the magnitude $||\nabla f||$ of the gradient and the orientation $\varphi(\nabla f)$
- The gradient magnitude gives the amount of the difference between pixels in the neighbourhood which gives the strength of the edge
- The gradient magnitude is defined by

$$|\nabla f| = \left\| \begin{bmatrix} G_x \\ G_y \end{bmatrix} \right\| = \left\| \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix} \right\| = \left[G_x^2 + G_y^2 \right]^{\frac{1}{2}}$$

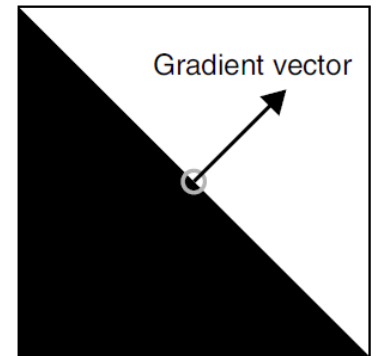
- The magnitude of the gradient gives the maximum rate of increase of $f(x, y)$ per unit distance in the gradient orientation of $|\nabla f|$

Gradient Operator

- The gradient orientation gives the direction of the greatest change, which presumably is the direction across the edge
- The gradient orientation is given by

$$\phi(\nabla f) = \tan^{-1} \left(\frac{G_y}{G_x} \right)$$

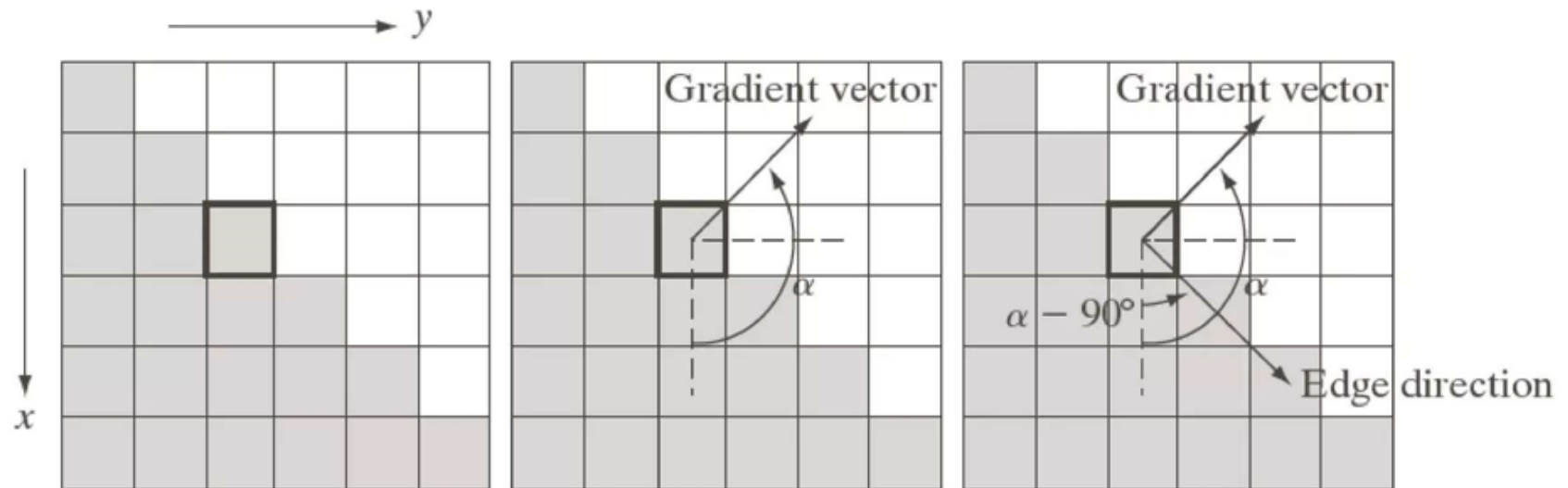
- where the angle is measured with respect to the x-axis



Gradient of edge pixel

- The direction of the edge at (x, y) is perpendicular to the direction of the gradient vector at that point
- An edge pixel is described using two important features:
 - (i) Edge strength, which is equal to the magnitude of the gradient
 - (ii) Edge direction, which is equal to the angle of the gradient

Gradient Operator



a b c

FIGURE 10.12 Using the gradient to determine edge strength and direction at a point. Note that the edge is perpendicular to the direction of the gradient vector at the point where the gradient is computed. Each square in the figure represents one pixel.

Gradient Operator

Roberts cross-gradient operators →

-1	0	0	-1
0	1	1	0

Roberts

Prewitt operators →

-1	-1	-1	-1	0	1
0	0	0	-1	0	1
1	1	1	-1	0	1

Prewitt

Sobel operators →

-1	-2	-1	-1	0	1
0	0	0	-2	0	2
1	2	1	-1	0	1

Sobel

Gradient Operator

Prewitt masks for
detecting diagonal edges



0	1	1	-1	-1	0
-1	0	1	-1	0	1
-1	-1	0	0	1	1

Prewitt

Sobel masks for
detecting diagonal edges



0	1	2	-2	-1	0
-1	0	1	-1	0	1
-2	-1	0	0	1	2

Sobel

a	b
c	d

FIGURE 10.9 Prewitt and Sobel masks for detecting diagonal edges.

Gradient Operator – An Example

a	b
c	d

FIGURE 10.10

(a) Original image. (b) $|G_x|$, component of the gradient in the x -direction. (c) $|G_y|$, component in the y -direction. (d) Gradient image, $|G_x| + |G_y|$.

$$\nabla f \approx |G_x| + |G_y|$$



Gradient Operator – An Example



a	b
c	d

FIGURE 10.11
Same sequence as
in Fig. 10.10, but
with the original
image smoothed
with a 5×5
averaging filter.

Gradient Operator – An Example



a b

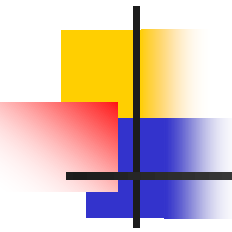
FIGURE 10.12
Diagonal edge
detection.

(a) Result of using
the mask in
Fig. 10.9(c).

(b) Result of using
the mask in
Fig. 10.9(d). The
input in both cases
was Fig. 10.11(a).

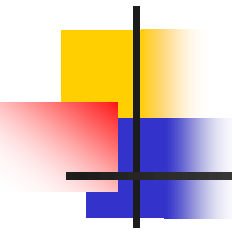
0	1	2
-1	0	1
-2	-1	0

-2	-1	0
-1	0	1
0	1	2



Second-Derivative Method of Detecting Edges in an Image

- Finding the ideal edge is equivalent to finding the point where the derivative is maximum or minimum
- The maximum or minimum value of a function can be computed by differentiating the given function and finding places where the derivative is zero
- Differentiating the first derivative gives the second derivative
- Finding the optimal edges is equivalent to finding places where the second derivative is zero
- The differential operators can be applied to images; the zeros rarely fall exactly on a pixel Typically, they fall between pixels
- The zeros can be isolated by finding the zero crossings
- Zero crossing is the place where one pixel is positive and a neighbouring pixel is negative



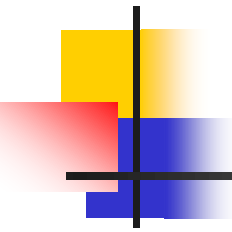
Second-Derivative Method of Detecting Edges in an Image

- The problems with zero-crossing methods are the following:
 - I. Zero crossing methods produce two-pixel thick edges
 - II. Zero crossing methods are extremely sensitive to noise
- For images, there is a single measure, similar to the gradient magnitude, that measures the second derivative, which is obtained by taking the dot product of ∇ with itself

$$\nabla \cdot \nabla = \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{bmatrix} \cdot \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{bmatrix}$$

$$\nabla \cdot \nabla = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

- The operator $\nabla \cdot \nabla = \nabla^2$ is called the Laplacian operator



Second-Derivative Method of Detecting Edges in an Image

- The Laplacian operator when applied to the function f , we get

$$\nabla^2 f = \left(\begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{bmatrix} \cdot \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{bmatrix} \right) f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

- The Laplacian operation can be expressed in terms of difference equations as given below:

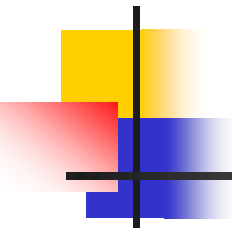
$$\frac{\partial f}{\partial x} = f(x+1, y) - f(x, y)$$

- and

$$\frac{\partial^2 f}{\partial x^2} = f(x+1, y) - 2f(x, y) + f(x-1, y)$$

- also

$$\frac{\partial^2 f}{\partial y^2} = f(x, y+1) - 2f(x, y) + f(x, y-1)$$



Second-Derivative Method of Detecting Edges in an Image

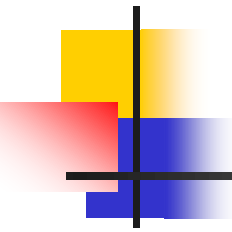
- This implies that

$$\nabla_f^2 = [f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1)] - 4f(x, y)$$

- The 3×3 Laplacian operator is given by

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

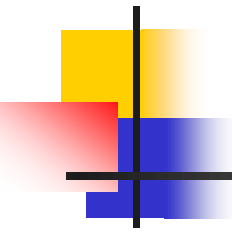
- The Laplacian operator subtracts the brightness values of each of the neighbouring pixels from the central pixel
- When a discontinuity is present within the neighbourhood in the form of a point, line or edge, the result of the Laplacian is a non-zero value
- It may be either positive or negative depending where the central point lies with respect to the edge



Second-Derivative Method of Detecting Edges in an Image

- The Laplacian operator is rotationally invariant
- The Laplacian operator does not depend on direction as long as they are orthogonal

- **Drawbacks of the Laplacian Operator:**
 1. Useful directional information is not available by the use of a Laplacian operator
 2. The Laplacian, being an approximation to the second derivative, doubly enhances any noise in the image



Laplacian of Gaussian (LOG)

- A prominent source of performance degradation in the Laplacian operator is noise in the input image
- The noise effects can be minimised by smoothing the image prior to edge enhancement
- The Laplacian-of-Gaussian (LOG) operator smooths the image through convolution with a Gaussian-shaped kernel followed by applying the Laplacian operator

Detection of Discontinuities Gradient Operators

- Consider the function:

$$h(r) = -e^{-\frac{r^2}{2\sigma^2}} \quad \text{where } r^2 = x^2 + y^2$$

and σ : the standard deviation

A Gaussian function

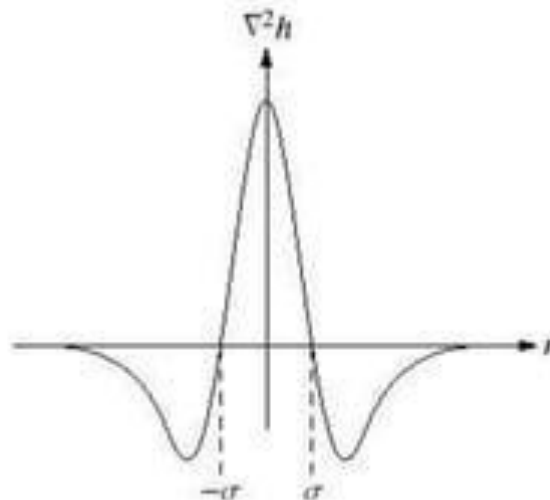
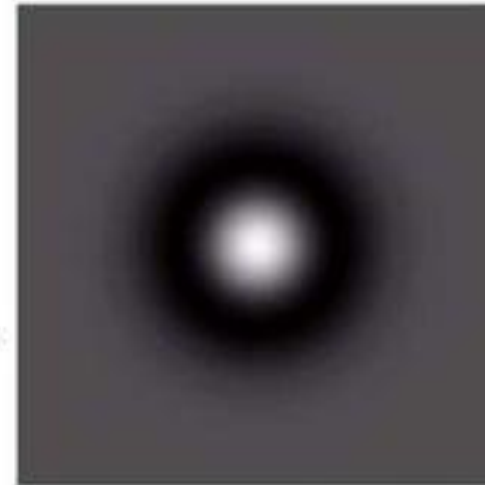
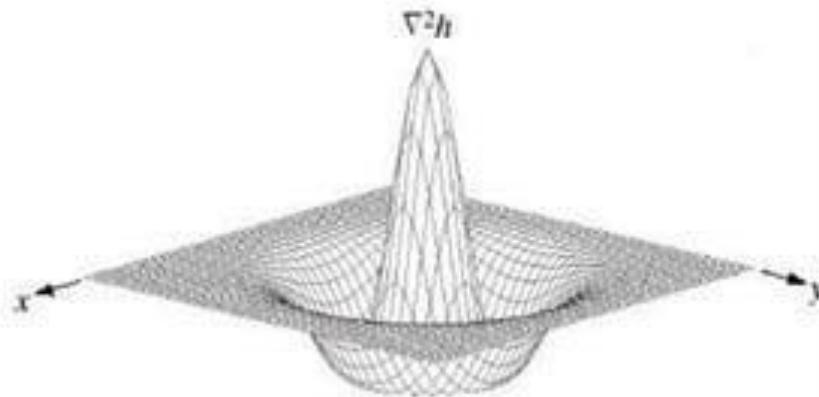
- The Laplacian of h is

$$\nabla^2 h(r) = -\left[\frac{r^2 - \sigma^2}{\sigma^4} \right] e^{-\frac{r^2}{2\sigma^2}}$$

The Laplacian of a
Gaussian (LoG)

- The Laplacian of a Gaussian sometimes is called the **Mexican hat function**. It also can be computed by **smoothing the image with the Gaussian smoothing mask, followed by application of the Laplacian mask.**

Detection of Discontinuities Gradient Operators



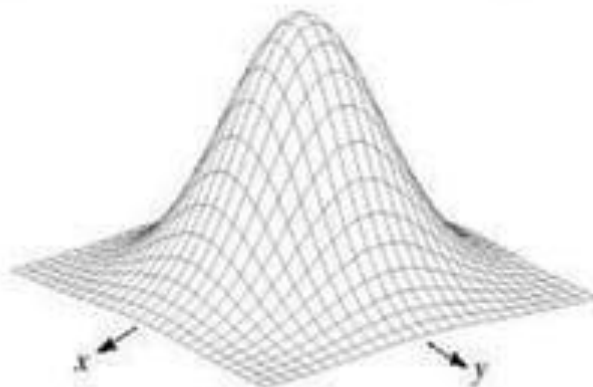
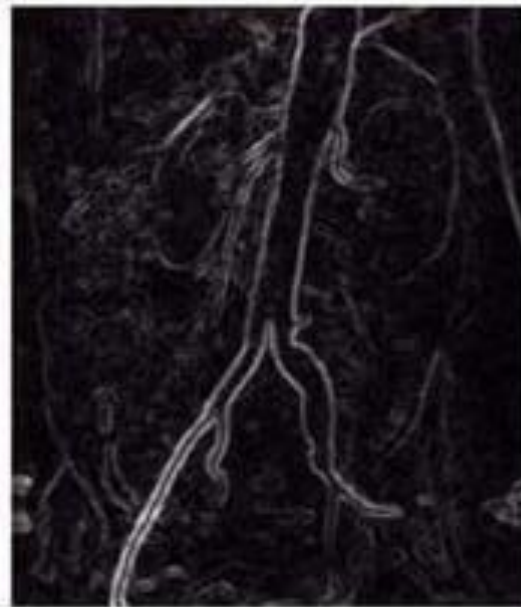
0	0	-1	0	0
0	-1	-2	-1	0
-1	-2	16	-2	-1
0	-1	-2	-1	0
0	0	-1	0	0

a b
c d

FIGURE 10.14
Laplacian of a Gaussian (LoG).
(a) 3-D plot.
(b) Image (black is negative, gray is the zero plane, and white is positive).
(c) Cross section showing zero crossings.
(d) 5×5 mask approximation to the shape of (a).

Detection of Discontinuities

Gradient Operators: Example



Sobel gradient

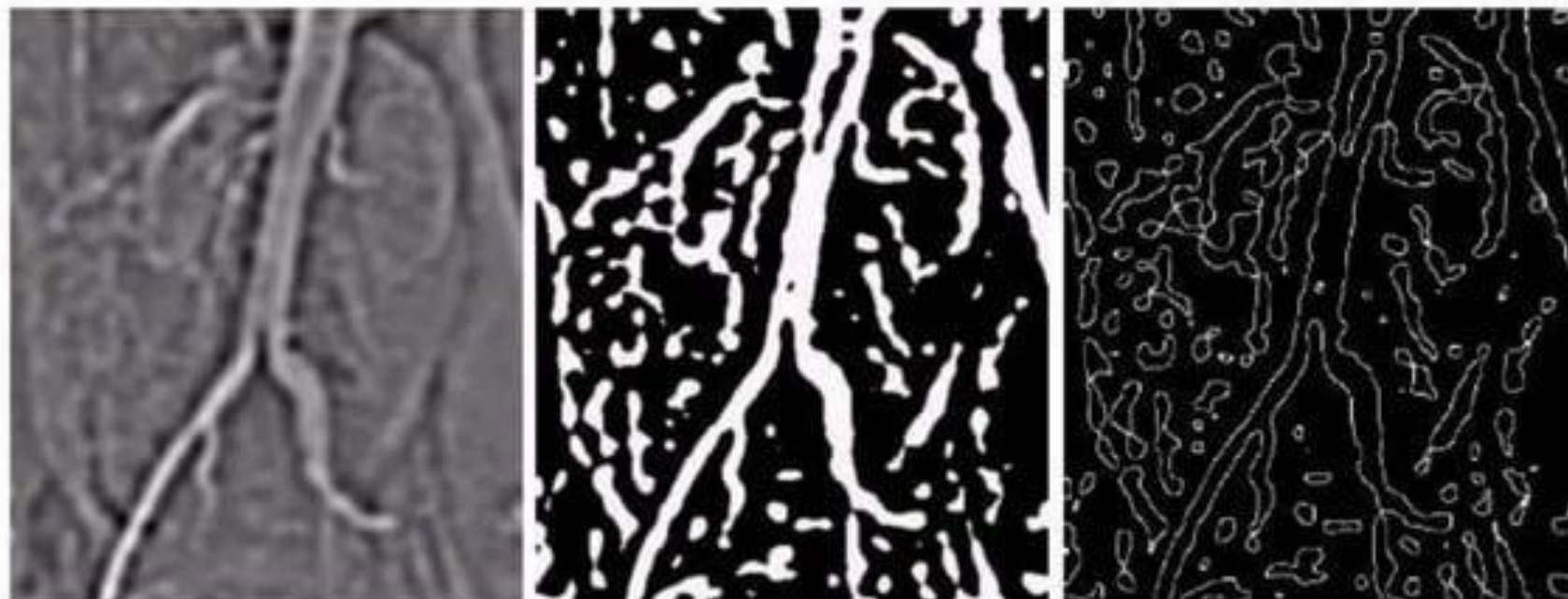
-1	-1	-1
-1	8	-1
-1	-1	-1

Gaussian smooth function

Laplacian mask

Detection of Discontinuities

Gradient Operators: Example



a b
c d
e f g

FIGURE 10.15 (a) Original image. (b) Sobel gradient (shown for comparison). (c) Spatial Gaussian smoothing function. (d) Laplacian mask. (e) LoG. (f) Thresholded LoG. (g) Zero crossings. (Original image courtesy of Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)

Edge Linking and Boundary Detection

Local Processing

- Two properties of edge points are useful for edge linking:
 - the strength (or **magnitude**) of the detected edge points
 - their **directions** (determined from gradient directions)
- This is usually done in **local neighborhoods**.
- Adjacent edge points with **similar** magnitude and direction are linked.
- For example, an edge pixel with coordinates (x_0, y_0) in a predefined neighborhood of (x, y) is similar to the pixel at (x, y) if
$$|\nabla f(x, y) - \nabla f(x_0, y_0)| \leq E, \quad E: \text{a nonnegative threshold}$$
$$|\alpha(x, y) - \alpha(x_0, y_0)| < A, \quad A: \text{a nonnegative angle threshold}$$

Edge Linking and Boundary Detection

Local Processing: Algorithm

1. Compute the gradient magnitude and angle arrays, $M(x, y)$ and $\alpha(x, y)$, of the input image, $f(x, y)$.
2. Form a binary image, $g(x, y)$, whose value at any pair of coordinates (x, y) is given by:

$$g(x, y) = \begin{cases} 1, & M(x, y) > T_M \text{ AND } \alpha(x, y) = A \pm T_A \\ 0, & \text{Otherwise} \end{cases}$$

where T_M is a threshold, A is a specific angle direction, and $\pm T_A$ defines a “band” of acceptable direction about A .

3. Scan the rows of g and fill all gaps in each row that do not exceed a specified length, K .
4. To detect gaps in other direction, θ , rotate g by this angle and apply the horizontal scanning procedure in step 3. Rotate the result back by $-\theta$.

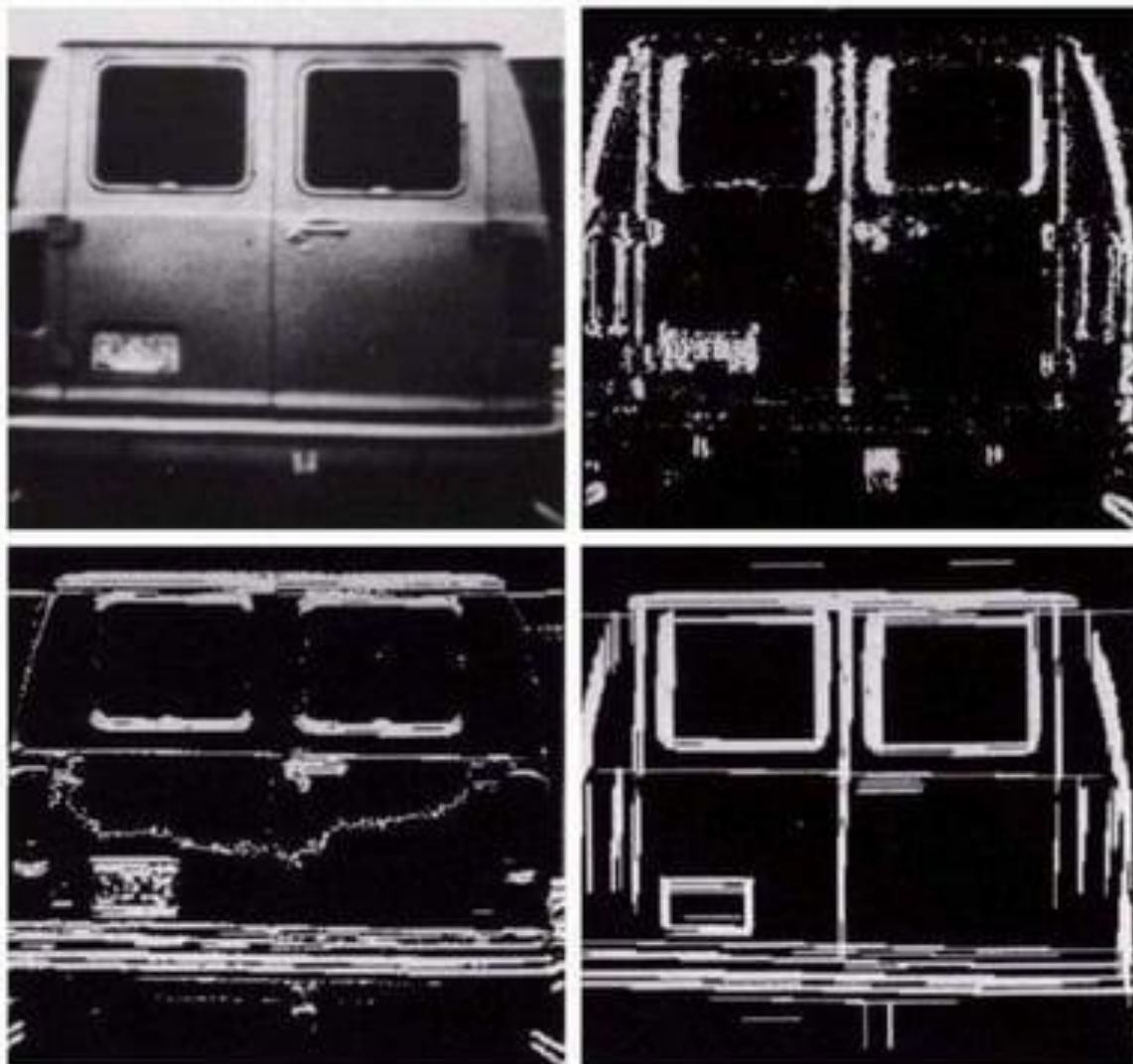
Edge Linking and Boundary Detection

Local Processing: Example

a b
c d

FIGURE 10.16

(a) Input image.
(b) G_y component
of the gradient.
(c) G_x component
of the gradient.
(d) Result of edge
linking. (Courtesy
of Perceptics
Corporation.)



In this example,
we can find the
license plate
candidate after
edge linking
process.

Edge Linking and Boundary Detection

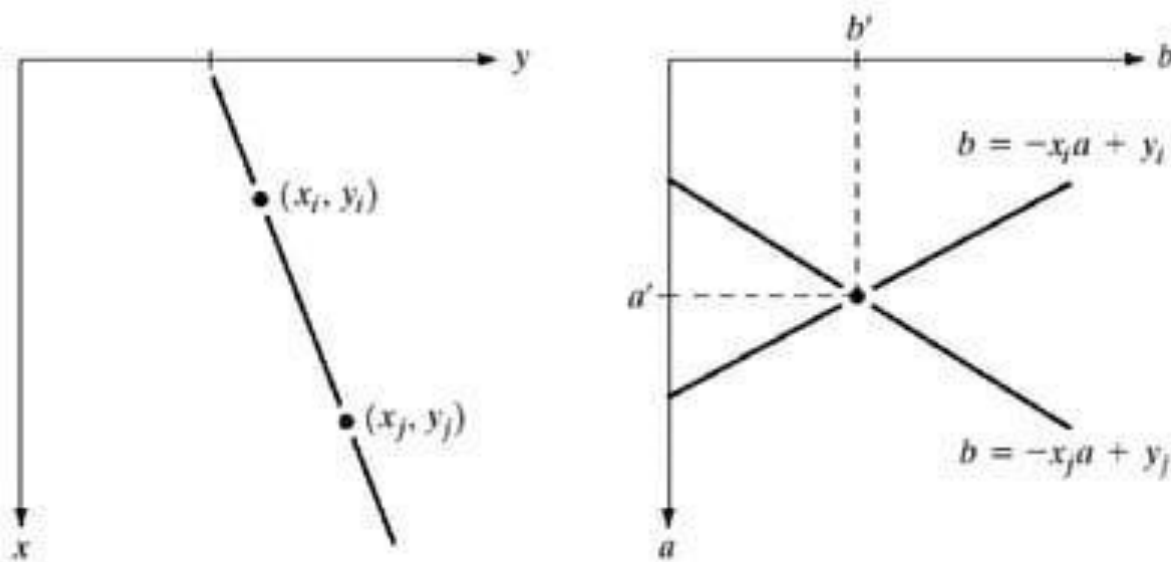
Global Processing via the Hough Transform

Edge Linking and Boundary Detection

Global Processing via the Hough Transform

- Hough transform: a way of finding edge points in an image that lie along a straight line.
- Example: xy -plane v.s. ab -plane (parameter space)

$$y_i = ax_i + b$$

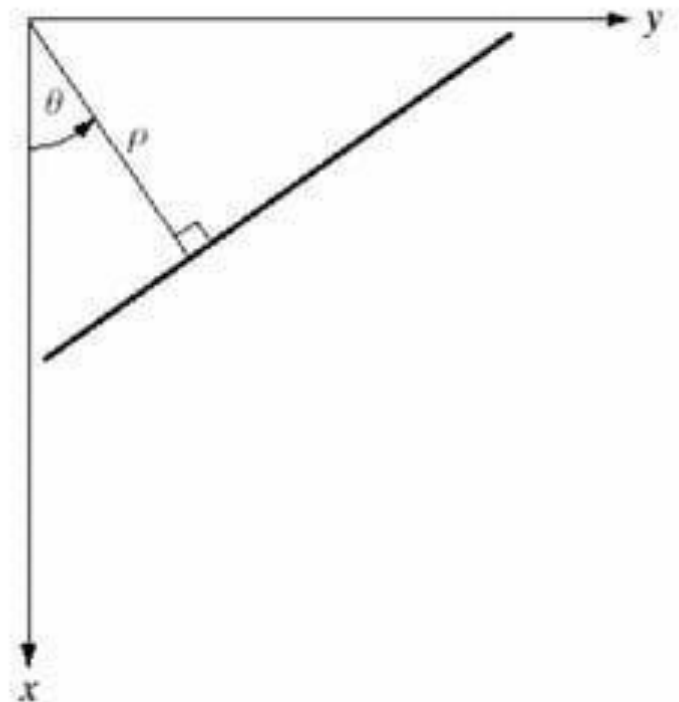


a b

FIGURE 10.17
(a) xy -plane.
(b) Parameter space.

Edge Linking and Boundary Detection Global Processing via the Hough Transform

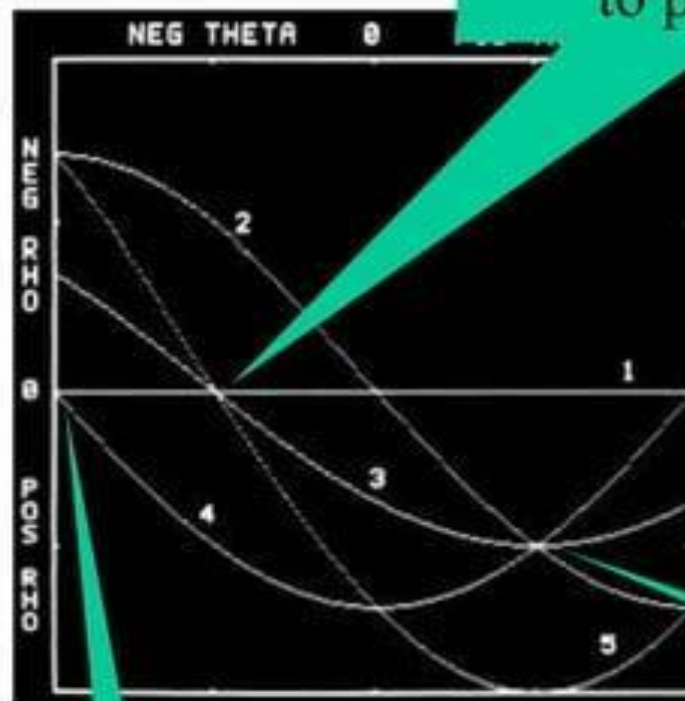
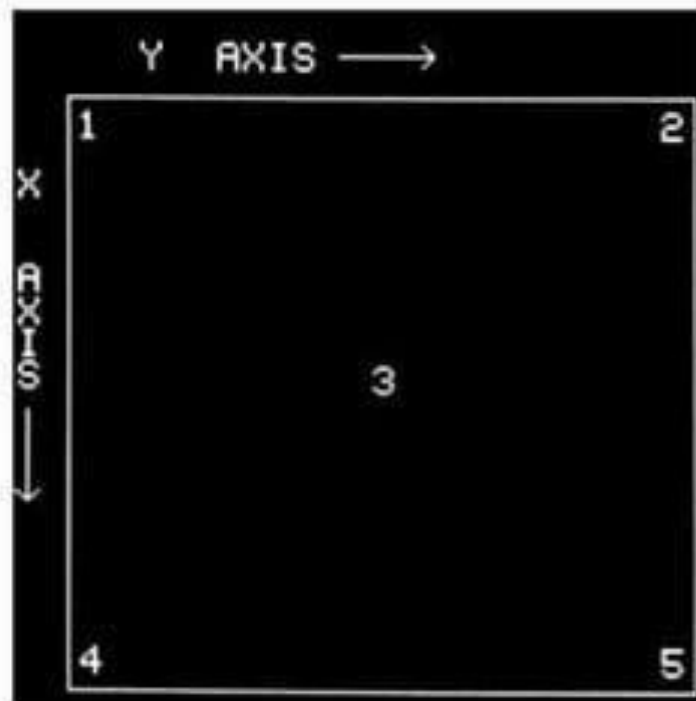
- The Hough transform consists of finding all pairs of values of θ and ρ which satisfy the equations that pass through (x,y) .
- These are accumulated in what is basically a 2-dimensional histogram.
- When plotted these pairs of θ and ρ will look like a **sine** wave. The process is repeated for all appropriate (x,y) locations.



$$x \cos \theta + y \sin \theta = \rho$$

Edge Linking and Boundary Detection

Hough Transform Example

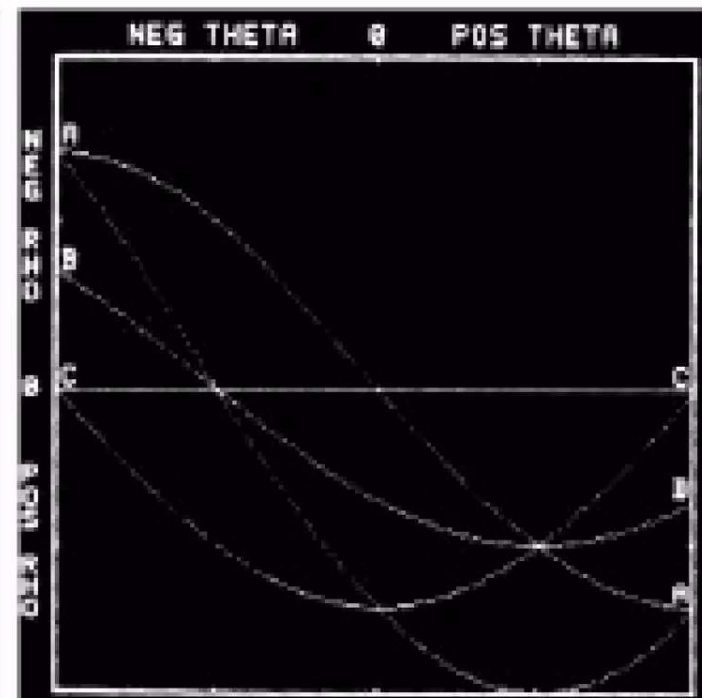
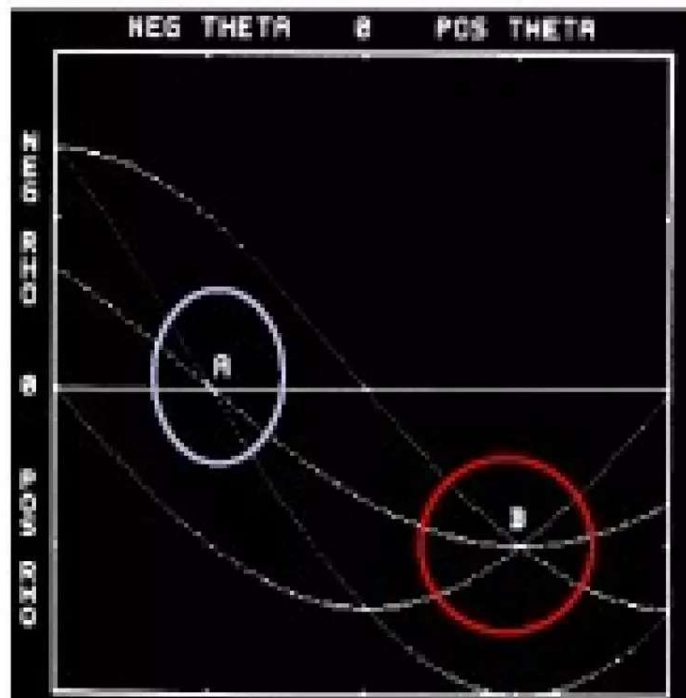
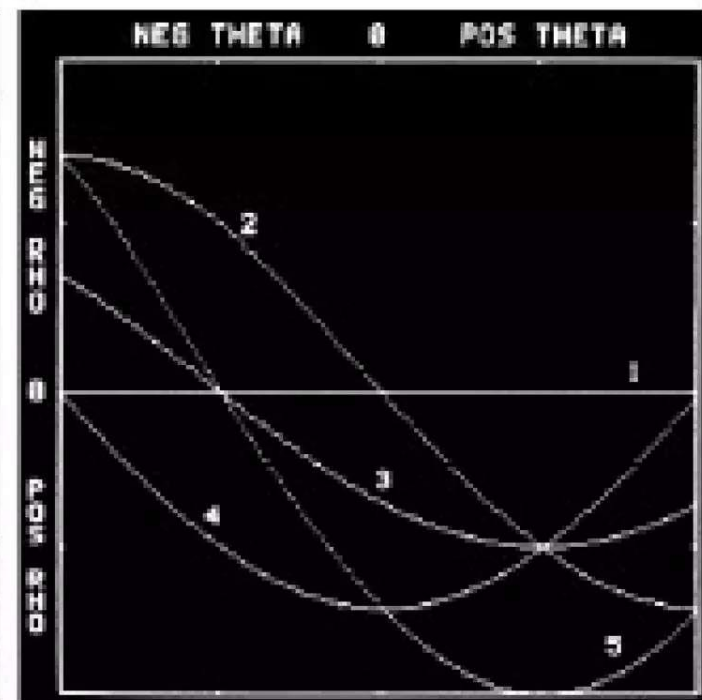
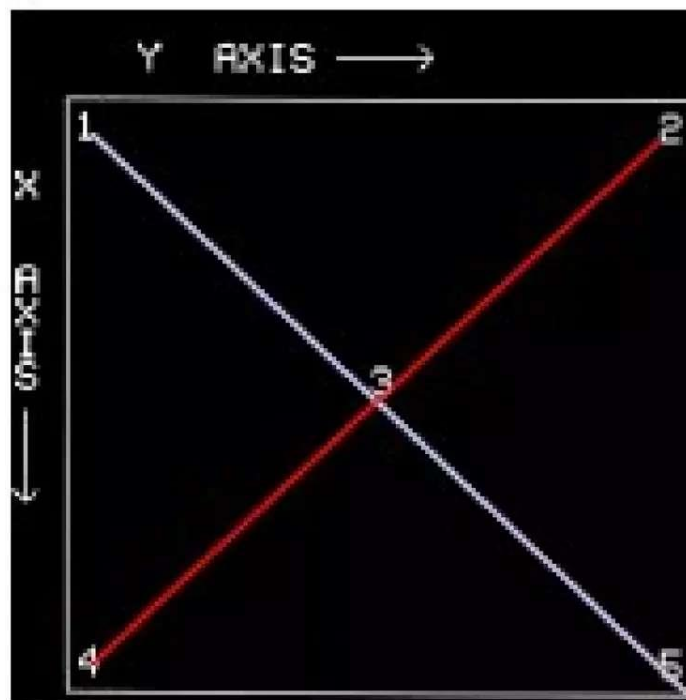


The intersection of the curves corresponding to points 1,3,5

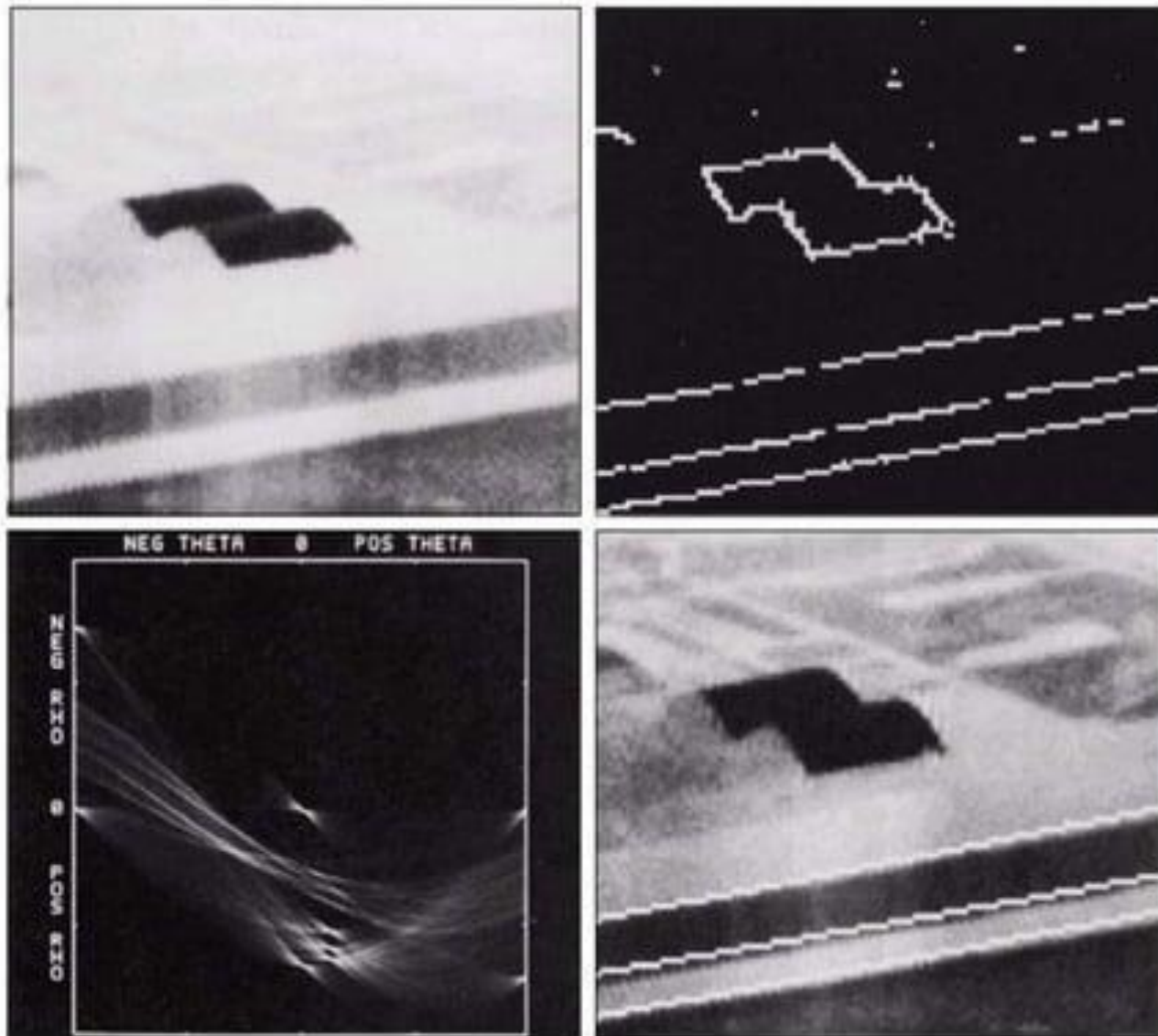
FIGURE 10.20
Illustration of the Hough transform.
(Courtesy of Mr. D. R. Cate, Texas Instruments, Inc.)

1,4

2,3,4



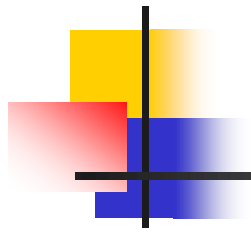
Edge Linking and Boundary Detection Hough Transform Example



a b
c d

FIGURE 10.21

(a) Infrared image.
(b) Thresholded gradient image.
(c) Hough transform.
(d) Linked pixels.
(Courtesy of Mr. D. R. Cate, Texas Instruments, Inc.)



THANK YOU